External-FDT: Cloud-to-Building Control System Tunneling in BACnet Networks

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Abstract

Building automation systems increasingly bridge private facility networks to public cloud infrastructure, creating undocumented attack surfaces. Through longitudinal analysis of internet-exposed BACnet Broadcast Management Devices (BBMDs), we discover a systemic pattern we term **External-FDT**: 71% of publicly visible BBMDs maintain Foreign Device Table entries pointing to public, routable IP addresses. Of these, 42% tunnel to major cloud providers (AWS, Azure, DigitalOcean). We also discover a novel **integrator concentration** phenomenon where a single external endpoint serves multiple distinct building networks. Our empirical analysis of 17 seeded BBMDs over 47 days exposes potential supply-chain attack surfaces in building automation infrastructure. All findings derive from passive OSINT analysis.

1 Introduction

Building automation systems—responsible for HVAC, access control, and critical facility operations—have historically relied on isolated networks. The BACnet protocol enables these systems to span subnets through Broadcast Management Devices (BBMDs) that forward traffic to registered Foreign Devices.

We discover that the majority of internet-exposed BBMDs maintain Foreign Device registrations with **public, routable IP addresses**—creating undocumented bridges to external infrastructure including major cloud providers.

1.1 Contributions

This paper makes three empirical contributions:

1. **External-FDT characterization**: We quantify the prevalence (71%) and patterns of public-IP Foreign Device registrations.
2. **Integrator Concentration**: We identify a novel pattern where a single external endpoint serves multiple distinct building networks.
3. **Attribution methodology**: We validate passive OSINT methods for identifying cloud-provider infrastructure without active probing.

2 Background

BACnet BBMDs enable cross-subnet communication by forwarding broadcasts to registered Foreign Devices. A Foreign Device Table entry contains: IP address, port, and TTL.

2.1 Related Work

Prior ICS research focused on protocol vulnerabilities and port exposure. No prior work has characterized tunneling relationships between building control systems and external infrastructure.

3 Methodology

Ethical constraints: All analysis uses passive OSINT only—Shodan-indexed banners. No active connection attempts.

Data: Longitudinal Shodan dataset: 17 seeded BBMDs, March–April 2026 (47 days).

Classification: We categorize FD IPs as Internal (RFC1918) or External (public routable), with External further classified by cloud provider attribution.

4 Results

4.1 Prevalence

Of 17 analyzed BBMDs, 12 (71%) exhibited External-FDT behavior.

4.2 Cloud Distribution

Provider	**Count**
DigitalOcean	4
Amazon AWS	2
Microsoft Azure	1
Other / Unknown	5

Table 1: Cloud Provider Attribution

DigitalOcean shows unexpected dominance (33%).

4.3 Integrator Concentration

Two distinct BBMDs share a common Foreign Device:

Attribute	**Value**
BBMD 1	66.58.248.125
BBMD 2	24.237.132.230
Shared FD	216.67.73.166
Separation	4,000 km
Duration	47 days continuous

Table 2: Integrator Concentration Pattern

This suggests a single integrator monitoring station bridging multiple client facility networks—creating a supply-chain attack surface.

5 Implications

External-FDT creates three attack surfaces:

1. Cloud compromise yields building network access
2. Single integrator breach affects multiple facilities
3. Internet-exposed FD endpoints compound risk

6 Conclusion

We demonstrate that cloud-to-building BACnet tunneling is systemic. The integrator concentration pattern suggests novel supply-chain attack surfaces in critical infrastructure.

Acknowledgments

This research used only publicly available Shodan data. All findings reported to US-CERT for coordinated disclosure.